

ORIGINAL ARTICLE

Evolving strategies in prostate cancer: Emerging approaches and unmet needs from the Bridging the Gaps in Prostate Cancer expert panel

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Abstract

Background: The expansion of treatment options for prostate cancer (PC) has improved disease-specific and overall survival outcomes but has also raised questions about the optimal level of treatment needed for patients based on their individual prognosis and accounting for potential toxicity, incorporating quality of life considerations.

Methods: A panel of experts met to discuss current controversies in the care of patients with PC across the disease continuum. Multidisciplinary experts review advances and persistent uncertainties in biomarker-guided assessment, imaging, and systemic therapy for prostate cancer. The discussion outlines priority gaps in evidence that must be addressed to optimize individualized patient care.

Results: Workshop topics included use of genomic biomarkers and artificial intelligence-guided tools to identify and manage high-risk and very-high risk localized disease, management of biochemical recurrence, identification of patients with metastatic hormone-sensitive PC who warrant treatment escalation, radiopharmaceutical therapy for metastatic castration-resistant PC including optimal sequencing of approved therapies, role of imaging in identification and management of extraprostatic disease, and lifestyle interventions to optimize survivorship.

Conclusions: Many questions remain about management of PC related to biomarker-based risk stratification to guide treatment selection, use of prostate-specific membrane antigen-positron emission tomography, and balancing the risk for PC-related death with risks for treatment-related toxicity. Ongoing research efforts are needed to optimize risk-based treatment, sequence of therapies throughout the disease continuum, and survivorship care.

KEYWORDS

androgen deprivation therapy, androgen receptor pathway inhibitors, biomarkers, prostate cancer treatment, PSMA-PET imaging, quality of life, radiopharmaceutical therapy, survivorship

INTRODUCTION

Management of prostate cancer (PC) has evolved rapidly with the development of treatment combinations based on presumed risk for disease progression, but gaps remain in this risk-based treatment approach. The “Bridging the Gaps (BTG) in Prostate Cancer” meeting was held on March 27, 2025, in Weehawken, New Jersey and included a diverse group of PC experts who practice in academic, community, and veterans health care system settings in medical oncology, urology, and radiation oncology. These panelists discussed emerging evidence, areas of consensus, and challenges across several areas of the PC care continuum, including high-risk localized prostate cancer, biochemical recurrence (BCR), metastatic hormone-sensitive prostate cancer (mHSPC), metastatic castration-resistant prostate cancer (mCRPC), and complications associated with therapy. This review summarizes a collective analysis, evaluation, and opinion of

the expert panel on the identified gaps and controversies in the treatment and management of PC, as well as the management of treatment complications. Table 1 provides an overview of key knowledge gaps the panel identified at the meeting.

RESULTS AND DISCUSSION**High-risk localized prostate cancer**

In the National Comprehensive Cancer Network (NCCN) guidelines, classification of high-risk and very high-risk localized PC focuses on clinicopathologic features.¹ Patients with high- or very high-risk localized PC exhibit heterogeneous outcomes, underscoring the need for strategies to improve identification of patients who are likely to benefit from treatment escalation (or de-escalation).²

TABLE 1 Overview of key knowledge gaps.

High-risk localized prostate cancer

- Effects of high interobserver variability of DRE and multiparametric MRI and accuracy of risk classification
- Utility of genomic biomarkers for prognostication and clinical decision-making
- Escalation and de-intensification strategies for patients with localized disease receiving definitive treatment
- Long-term effects of moving therapies earlier in the treatment sequence
- False-positive results on PSMA-PET
- Clinical implications and risks of overtreatment of PSMA-positive, conventional imaging-negative micrometastatic lesions
- Inability to rule out need for pelvic lymph node dissection with PSMA-PET
- Effect of ADT (and potentially ARPIs) on performance characteristics of PSMA-PET

Metastatic hormone-sensitive prostate cancer

- Role of PSMA-PET in defining high-volume disease
- Relevance of docetaxel with the increased use of targeted therapies
- Better selection of patients enrolled in biomarker-guided clinical trials
- Increasing uptake of germline and tumor genetic testing in clinical practice
- Whether increasing uptake of PSMA PET increases risk for overtreatment
- Optimal use of prostate- and metastasis-directed radiotherapy for de novo, low-volume disease
- Early and late radiotherapy-related toxicities

Metastatic castration-resistant prostate cancer

- Optimal dose and fractionation of PSMA-targeted RLT
- Safety of concomitant therapies with PSMA-targeted RLT
- Efficacy and safety of intermittent PSMA-targeted RLT
- Utility of PSMA PET imaging to select patients for PSMA-targeted RLT, individualize dosimetry, and monitor response and adjust dosing accordingly
- Defining the role of ^{223}Ra combined with enzalutamide in patients who are refractory to an earlier ARPI
- Evaluation of treatment response to ^{223}Ra treatment
- Relationship between exposure to PARP inhibitors and resistance to ^{177}Lu -PSMA-617
- Long-term bone marrow toxicity with ^{177}Lu -PSMA-617
- Effective treatment of AR amplifications
- Utility of novel AR-targeted agents before the mCRPC setting (e.g., mHSPC)

Complications of systemic therapy

- Benefits of BSAs in the mHSPC setting

(Continues)

TABLE 1 (Continued)

- Methods to implement BSAs in medical, radiation, and urologic oncology clinics and collaborating with endocrinologists for patients at highest risk for SREs
- Implementation of CVD risk assessment in patients with PC
- Develop educational content about strategies to mitigate CVD risk
- Identification of barriers to CV care
- Implement health technology to improve assessment of and access to cardiovascular and PC care
- Differentiation between cognitive impairment and/or depression and fatigue related to treatment or sleep issues
- Link between androgen-targeted therapy and cognitive impairment and/or depression
- Strategies for lifestyle modification to optimize survivorship

Abbreviations: ^{177}Lu -PSMA-617, lutetium-177 prostate-specific membrane antigen-617; ^{223}Ra , radium-223; ADT, androgen deprivation therapy; AR, androgen receptor; ARPI, androgen receptor pathway inhibitor; BCR, biochemical recurrence; BSA, bone-supporting agent; CV, cardiovascular; CVD, cardiovascular disease; DRE, digital rectal examination; HRR, homologous recombination repair; mCRPC, metastatic castration-resistant prostate cancer; mHSPC, metastatic hormone-sensitive prostate cancer; PARP, poly(ADP-ribose) polymerase; PC, prostate cancer; PSMA-PET, prostate-specific membrane antigen-positron emission tomography; RLT, radioligand therapy; SRE, skeletal-related event.

Biomarkers

Clinicopathologic features alone are poor predictors of metastasis development within the NCCN-defined high-risk PC group, highlighting the need for additional biomarkers to predict which patients are likely to develop metastases and benefit from intensified therapy.³ A 22-gene genomic classifier (GC; DECIPHER) was validated in pre-treatment prostate biopsy samples, with higher GC scores independently associated with distant metastases, PC-specific mortality, and overall survival (OS).² Use of the DECIPHER GC with a digital pathology artificial intelligence model also significantly predicted distant metastasis in a prospective cohort of almost 10,000 patients with localized PC.⁴ Two NRG Oncology trials (PREDICT-RT [NCT04513717] and Guidance [NCT05050084]) are studying whether patients can receive intensified or deintensified treatment regimens based on their Decipher-GC scores with noninferior outcomes.

A multimodal artificial intelligence-derived biomarker test (Artera MMAI), trained using pretreatment digital prostate biopsy images and clinical data (age, PSA, Gleason score, and T stage) from six NRG Oncology phase 3 randomized radiotherapy trials, predicted benefit from long- versus short-term ADT in patients with high-risk nonmetastatic PC.⁵ Patients classified as MMAI biomarker positive had significantly lower rates of distant metastases with long-term ADT, whereas distant metastasis rates were not significantly different between long- and short-term ADT in those classified as MMAI biomarker negative.⁵ Researchers concluded that the tool

could be used to guide duration of ADT in the high-risk non-metastatic PC population.⁵

The BTG panelists agreed that identifying biomarkers up front in the localized PC setting might help identify the subset of patients who are most likely to benefit from intensified or novel therapies. However, more evidence is needed to support widespread use of genomic biomarkers for prognostication and clinical decision-making in high-risk localized PC and elucidate the long-term effects of moving therapies earlier in the treatment sequence.

Imaging for high-risk localized PC

Prostate-specific membrane antigen-positron emission tomography (PSMA-PET) has emerged as a highly sensitive modality for detecting occult metastatic disease in the initial staging and BCR settings and is the preferred imaging modality for the BTG panelists.¹ Multiple trials (e.g., OSPREY [NCT02981368] and proPSMA trial [NCT0357774]) demonstrated the superior performance of PSMA-PET compared with conventional imaging for detecting pelvic lymph node and metastatic disease in high-risk PC.^{6,7} Given its superior performance characteristics, PSMA-PET is now strongly recommended for staging high-risk patients in contemporary guidelines.¹ However, PSMA-PET's superior sensitivity for detecting occult metastatic disease compared to conventional imaging may lead to stage migration, with some patients previously classified as having localized disease now are being identified as having metastatic disease. Although foregoing curative intent therapy may not be in these patients' best interest, treatment plans should address PET detected metastases. Although prospective data support a survival benefit for primary lesion-directed therapy in patients with low-volume metastatic HSPC as defined by conventional imaging, whether these findings can be extrapolated to patients with PSMA-PET detected metastases that would have been occult on conventional imaging remains an area of active investigation.⁸ Ongoing studies such as INDICATE (EA8191) will provide data for whether early treatment with radiation to PET-detected oligometastases improves outcomes compared to long-term ADT with radiation to the prostate primary and pelvis.

The panelists also noted that the potential for false-positive results, particularly at bony sites, should be considered, and PSMA-PET findings should be assessed using a standardized reporting system (e.g., PRIMARY score, PROMISE, or PSMA-RADS) and interpreted in clinical context.⁹ PSMA-PET is considered only moderately sensitive for small lesions, and negative PSMA-PET imaging does not rule out pelvic lymph node disease therefore it is not recommended to omit pelvic lymph node dissection or radiation coverage of pelvic lymph nodes in this setting.⁶ Recent use of ADT has been shown to affect the performance characteristics of PSMA-PET; therefore, panelists agreed that clinicians should avoid the initiation ADT if they are considering PSMA-PET in the near future.¹⁰ More research is needed to clarify whether different types of ADT and androgen receptor pathway inhibitors (ARPIs) have similar effects on the performance characteristics of PSMA-PET.

ADT treatment Intensification

Radiotherapy and 1.5–3 years of ADT are recommended for patients with high-risk localized PC who have a life expectancy greater than 5 years or are symptomatic (observation or monotherapy with ADT or radiotherapy can be considered for asymptomatic patients with shorter life expectancies).¹ Addition of abiraterone to ADT and radiation therapy is recommended for very high-risk localized PC based on findings that addition of abiraterone to ADT significantly improved metastasis-free survival and OS in a patient population that closely reflects the NCCN-defined very high risk PC.^{1,11} Multiple ongoing phase 3 trials, including ENZARAD (NCT02446444), PROTEUS (NCT03767244), ATLAS (NCT02531516), and ERADICATE (NCT04484818), are evaluating whether addition of an ARPI can improve time to BCR or distant metastasis in patients with high-risk localized PC.

However, the BTG panelists noted the high interobserver variability of digital rectal examination (DRE) and multiparametric MRI (the latter of which also has modest sensitivity for detecting extraprostatic disease) could affect accuracy of T staging and thus classification of high-risk and very high-risk localized PC. With the increasing use of multiparametric MRIs for T staging, the necessity of DRE for clinical T staging was debated among panelists. Panelists agreed that nonimaging-based risk factors, including PSA doubling time (PSADT), risk for mortality due to non-PC causes, and genomic risk classifiers, should be considered when deciding whether to escalate or deescalate therapy.

Management of BCR

BCR after radical prostatectomy occurs in approximately one-third of patients within 10 years of receiving definitive therapy and may be indicative of micrometastatic, local, regional, or distant disease.¹² The evaluation and management of BCR has evolved significantly with the increased use of PSMA-PET, which identifies occult metastatic disease that would not be detected on conventional imaging. This enhanced detection capability has important implications for treatment selection, as therapeutic options may differ based on the presence or absence of PSMA-PET-detected metastases in patients without evidence of metastatic disease on conventional imaging. The management of BCR remains challenging, as multiple treatment options exist, albeit with no established standard of care, for patients without detectable metastases. Shorter PSA doubling time (PSADT) is associated with increased risk for disease progression and PC-specific mortality, and these patients could benefit from escalation of treatment beyond ADT and/or salvage radiotherapy typically used.^{12,13} In the EMBARK trial (NCT02319837), enzalutamide with leuprolide yielded superior metastasis-free survival, longer time to PSA progression, and superior 8-year OS rate compared with leuprolide alone in patients with biochemically recurrent PC and PSADT <9 months; enzalutamide monotherapy did not yield superior OS compared with leuprolide monotherapy.^{13,14} Additionally, a greater proportion of patients in the

enzalutamide-treated groups were able to suspend treatment at 37 weeks (allowed if PSA levels were undetectable), which may reduce toxic effects that occur with long-term treatment.¹³ The panelists noted that the PSADT that defines high risk remains an ongoing question, with some participants suggesting that a PSADT of <6 months may be a more suitable cutoff for delineating high-risk disease. Additionally, the EMBARK trial authors noted that their results must be considered in the context of the changing imaging landscape that involves increasing use of PSMA-PET, as the EMBARK participants did not have evidence of metastasis on conventional imaging but may have had disease that would have been detectable on PSMA-PET.¹⁴ In a patient cohort eligible for the EMBARK trial (i.e., patients with high-risk, nonmetastatic, hormone-sensitive PC), 84% had positive PSMA-PET results, 46% had M1 disease, and 24% had five or more metastatic lesions, highlighting the importance of considering PSMA-PET results when treating this population.¹⁵

For patients without metastases on conventional imaging but with PSMA-PET-detected oligometastatic disease, metastasis-directed therapy (MDT) may provide opportunities for early intervention.¹⁶ In the EXTEND trial (NCT03599765), addition of metastasis-directed radiotherapy (MDT) to intermittent hormone therapy was associated with longer median PFS in patients with PC and five or fewer metastatic sites amenable to MDT.¹⁷ Pooled analysis of the STOMP (NCT01558427) and ORIOLE (NCT02680587) trials showed that MDT improved rPFS versus surveillance in oligometastatic BCR, particularly in patients with high-risk mutations.¹⁸ However, the 5-year OS remained high (85% in STOMP, with less than one-half of deaths attributed to PC), highlighting the relatively good cancer-related prognosis and the importance of considering competing risk for death when developing a treatment plan for this patient population.¹⁹ The panelists also noted that PSMA-PET may identify patients with occult metastatic disease that would not have been detected using conventional imaging, and risk for over-treatment must be considered.

mHSPC

Despite significant advances in treatment options for mHSPC, a substantial proportion of patients still receive ADT monotherapy in clinical practice, falling short of established standards of care and representing a major gap in optimal patient management.²⁰ Patients with mHSPC comprise a highly heterogeneous population, and optimal systemic treatments vary considerably.²¹ An ARPI (e.g., abiraterone, apalutamide, and enzalutamide) with ADT is considered the minimum standard of care (SOC) for high-volume mHSPC, as multiple trials showed improved OS with addition of an ARPI to ADT.^{1,22–24} Darolutamide was recently Food and Drug Administration-approved in combination with ADT for mHSPC based on the ARANOTE trial (NCT04736199) showing improvement in radiographic progression or death (rPFS) of 46% versus placebo plus ADT.²⁵

Triplet therapies comprised of an ARPI, ADT, and docetaxel were shown to delay disease progression and improve OS compared with

ADT-docetaxel doublet therapy in the PEACE-1 (NCT01957436) and ARASENS trial (NCT02799602).^{21,26,27} However, the toxicity of the triplet regimen is higher, and most of the patients in both trials had high-risk, synchronous disease.^{21,26,27} Findings from a systematic review and meta-analysis suggest that triplet therapy may be preferred for synchronous, high-volume disease, whereas ARPI-ADT doublet therapy may be more appropriate for metachronous, low-volume disease (no OS benefit was observed with triplet therapy in this population).²¹ Furthermore, treatment-related toxicity must be considered in the context of a patient's life expectancy and quality of life.²¹ The panelists also noted that the additive benefit of docetaxel to ADT and ARPI is unclear; the ongoing CCTG-SWOG Triple Switch (NCT06592924) and Alliance ASPIRE (NCT06931340) trials are comparing docetaxel, ADT, and ARPI to ADT plus ARPI in patients with mHSPC.

Although uptake of ARPI in clinical practice has increased in recent years, it is still suboptimal according to real-world data and varies considerably with patient age, comorbidities, metastatic burden, geographic region, and institutional setting.^{28,29} The panelists also noted that questions remain about the role of PSMA-PET in defining high-volume disease and the relevance of docetaxel as newer therapies (e.g., PARP inhibitors and/or lutetium-177 [¹⁷⁷Lu]-PSMA-617) enter the clinical space for mHSPC.

Personalizing treatment using biomarkers and genomic testing

Although clinical factors such as disease volume and disease onset (de novo vs. metachronous) are currently used to guide selection of systemic treatment, biomarkers could help improve selection of treatment in mHSPC.³⁰ The first interim analysis of the AMPLITUDE trial (NCT04497844) showed improved rPFS and time to symptomatic progression with addition of niraparib (compared with placebo) to abiraterone and prednisone in patients with HRR-altered mHSPC (OS data were immature at the time of analysis).³⁰ According to the BTG panelists, these findings suggest promise for the use of HRR status to select patients for PARP inhibitor-based treatment, but additional biomarker-guided clinical trials are needed to determine its utility in informing clinical practice. Real-world data show that uptake of germline and tumor genetic testing is low, and most targetable HRR gene alterations are first identified in the mCRPC setting.³¹ This may be related to several factors, including variable clinician knowledge of testing guidelines, insufficient training in clinical genetics, limited clinic time for counseling, complex test ordering processes, patient cost concerns, and unclear roles about which practitioner should conduct testing.^{31,32}

Role of prostate-directed radiation therapy

Radiotherapy modalities have been explored as a way to reduce genitourinary symptoms, increase time to castration-resistant

disease and OS, and delay initiation of ADT in low-volume mHSPC.^{8,33} Addition of prostate-directed radiotherapy to abiraterone plus SOC (ADT with or without docetaxel) improved rPFS and castration resistance-free survival, reduced incidence of serious genitourinary events in patients with low-volume, de novo mHSPC in the PEACE-1 trial (NCT01957436), and improved 5-year OS when added to ADT monotherapy in patients with de novo, low-volume mHSPC in arm H of the STAMPEDE trial (NCT00268476).^{8,33} This research suggests that prostate-directed radiotherapy may provide patients with de novo, low-volume mHSPC, but panelists noted that more research is needed given the lack of OS benefit in PEACE-1. Early and late radiotherapy-related toxicities also remain a concern, particularly as the life expectancy of patients with PC increases.

Metastatic castration-resistant prostate cancer

PSMA-targeted radioligand therapy

Because most mCRPC lesions express PSMA, PSMA-targeted radioligand therapies (RLTs), which deliver beta particle radiation to PSMA-positive cells and the tumor microenvironment, are a promising strategy.³⁴ ¹⁷⁷Lu vipivotide tetraxetan (¹⁷⁷Lu-PSMA-617) was initially approved for mCRPC previously treated with ARPI and taxane-based chemotherapy, and was recently expanded to include patients with mCRPC who had received one ARPI and were eligible to delay taxane-based chemotherapy based on findings from the PSMAfore trial (NCT04689828).^{35–37} In this study, patients who received ¹⁷⁷Lu-PSMA-617 had a superior rPFS compared with those who received a change in ARPI therapy.³⁸ ¹⁷⁷Lu-PNT2002, another PSMA RLT, demonstrated superior rPFS and health-related quality of life versus ARPI change in the SPLASH trial (NCT04647526).³⁸ However, the relative efficacy of PSMA RLT versus docetaxel remains an unanswered question, as neither trial compared the two directly. Furthermore, patient selection criteria varied considerably among the RLT trials, and clarifying the clinical scenario in which RLT is most likely to be beneficial will be important in future research. Panelists also noted ongoing questions regarding optimal PSMA-targeted RLT dosing, fractionation, concomitant therapies, and intermittent scheduling. Additional research is needed on PSMA-PET for patient selection and dosimetry individualization.

Radium-223

The use of radium-223 (²²³Ra), a bone-targeting radiopharmaceutical that delivers high-energy alpha particles, for mCRPC with bone metastases offers overall survival benefit even if treatment is not often associated with significant PSA declines.³⁹ After its regulatory approval, subsequent data (ERA-223; NCT02043678) had raised clinical concerns that indicated the treatment was associated with increased risk of bone fractures.⁴⁰ Post hoc analysis revealed that use of bone-supporting agents (BSAs; e.g., denosumab, zoledronic

acid) was less frequent among patients who sustained a fracture.⁴⁰ More recently, a significant improvement in rPFS and OS at interim analysis, with no increase in fracture risk, was observed with the addition of ²²³Ra to enzalutamide in the PEACE-3 trial (NCT02194842).⁴¹ Notably, most patients in this trial were using BSAs after a trial amendment mandated the use of BSAs, in response to the learnings from the ERA-223 trial.

The panelists noted that ²²³Ra could be a particularly valuable option for targeting PC that does not express PSMA and bone-only disease. Whether ²²³Ra and ARPI had additive or synergistic effects is unclear from current PEACE-3 data, and the effectiveness of ²²³Ra in patients who are refractory to prior ARPIs should be investigated further, because only 2% of patients in the PEACE-3 trial received prior abiraterone.⁴¹

Sequencing of PARP inhibitors

Combining PARP inhibitors (PARPi), which had initially been used as monotherapy for mCRPC harboring HRR gene mutations, with ARPIs has been of interest because of synergy between the agents observed in the preclinical setting.^{42,43} Three PARPi-ARPI combinations (olaparib-abiraterone, talazoparib-enzalutamide, and niraparib-abiraterone) have been approved for this patient population (although OS benefit has only been demonstrated with the talazoparib-enzalutamide combination in the TALAPRO-2 trial [NCT03395197]).^{44–47} Panelists noted that the clinical effect of using PARPi and ARPIs sequentially remains an ongoing question, as subsequent use of PARPi is low in the control arms of the PARPi-ARPI combination trials. In the phase 2 BRCAaway trial (NCT03012321), concomitant treatment with olaparib and abiraterone was associated with longer progression-free survival than either agent given alone or in sequence among patients with mCRPC harboring *BRCA1/2* or *ATM* alterations.⁴⁸ Although these findings may suggest a clinical benefit of using the agents at the same time, the study was underpowered to address this question.⁴⁸ However, olaparib improved rPFS over an ARPI switch in taxane-naïve and taxane-exposed patients with mCRPC harboring *BRCA* alterations that had progressed on an ARPI in an exploratory subgroup analysis of the PROFOUND trial (NCT02987543), suggesting that sequencing PARPi after ARPI may still provide a clinical benefit.⁴⁹ This was further supported by an exploratory subgroup analysis of findings from the TRITON-3 trial (NCT02975934), in which imaging-based PFS was longer in patients with mCRPC harboring a *BRCA* or *ATM* alteration who received rucaparib compared with those who received docetaxel or ARPI switch, respectively.⁵⁰ However, panelists noted that small sample sizes limit applicability of findings from all these trials, and prospective longer-term follow-up with larger numbers of participants are needed to corroborate these findings and assess for OS outcomes. Additionally, use of the PARPi-ARPI combination may increasingly move into the mHSPC space in the future, if the positive findings from the AMPLITUDE trial continue with longer-term follow-up, which would introduce the question of which treatment

to choose after progression on this combination. The panelists also highlighted the need to study the relationship between prior use of PARPis and decreased response to ^{177}Lu -PSMA-617 in patients with DNA damage repair gene mutations, which has been observed in retrospective data.⁵¹

Novel AR therapies

The AR is a key mechanism of resistance in mCRPC through somatic mutations, gene amplification, overexpression, alterations in coactivators and corepressors, posttranslational modifications, and cross-talk between AR and its signaling molecules.⁵² This has increased interest in identifying novel AR-targeted therapies for the patients with mCRPC who develop resistance to currently available ARPIs.⁵³ Proteolysis-targeting chimera (PROTAC) AR degraders (e.g., ARV-766 and BMS-986365), which target the AR for degradation by binding to the ligand-binding domain (LBD) of AR, have been associated with PSA declines and clinical activity in patients with mCRPC, particularly those harboring a LBD mutation.^{53,54} However, panelists noted that it may be less effective for other AR alterations, particularly AR amplifications, which are often resistant to current therapies.

Opevesostat inhibits cytochrome P450 11A1, the initial and rate-limiting enzyme involved in synthesis of all steroid hormones, has been shown to yield PSA declines in patients with heavily pretreated mCRPC that harbored LBD mutations, in a phase 1/2 trial.⁵⁵ However, adrenal insufficiency was a key safety finding and required short-term high-dose glucocorticoid treatment in 37% of patients. Panelists noted that although it is important to identify adrenal insufficiency promptly, it is often easily managed with glucocorticoid therapy.

Complications of systemic therapy

Bone health

Use of ADT is associated with an increased risk for bone fractures, which are an independent predictor of mortality in patients with PC.^{56,57} More than 90% of patients with mCRPC have bone metastasis which, in combination with ongoing ADT, can further increase risk for skeletal-related events (SREs) and associated disability, quality of life deterioration, and mortality.⁵⁸ Use of BSAs can reduce risk for SREs in patients with mCRPC, with denosumab associated with longer time to first SRE compared with zoledronic acid.^{59,60} However, use of BSAs does not appear to prolong OS in the mHSPC population.⁵⁹ According to the panelists, the ideal duration of BSAs, strategies to address cost concerns (e.g., alternative dosing schedules), and optimal dose for mHSPC remain ongoing questions. Additionally, regular assessment of bone mineral density (every 1–2 years for patients on ADT, according to NCCN guidelines) and lifestyle interventions that support bone health (e.g., calcium and vitamin D supplementation, tobacco cessation, alcohol moderation, and weight-

bearing exercise) also may help mitigate fracture risk in patients taking ADT.^{1,61}

Osteonecrosis of the jaw is a potentially serious complication associated with BSAs, with a cumulative rate of 11% observed in a real-world setting of patients with mCRPC (this rate decreased significantly over the study period with regular dental assessment).⁶² However, the study also showed an OS benefit with use of BSAs in patients with mCRPC and bone metastases, and panelists agreed that the overall benefits of BSAs often outweigh the risks in this population.⁶² They noted that implementing use of BSAs (and possibly collaborating with endocrinologists, if possible) for patients at highest risk can be useful when managing bone health in patients with PC.

ADTs and cardiovascular risk

Despite the association between use of androgen-directed therapies and increased risk for atherosclerotic cardiovascular disease (ASCVD), patients with PC are under evaluated and undertreated for ASCVD.^{63,64} Risk for major adverse cardiovascular events is also higher in Black patients with PC compared with White patients, with much of this disparity mediated by socioeconomic status.⁶⁵ The panelists noted that initiatives are needed to implement CVD risk assessment, develop educational content about strategies to mitigate CVD risk, and design studies that identify barriers to cardiovascular care and the implementation of healthy lifestyle behaviors and to optimally leverage health technology to improve assessment of and access to cardiovascular and PC care. The ongoing CARE-PC trial (NCT06064149) is testing a web-based, quality improvement tool to educate patients taking ADT for at least 6 months about CVD risk assessment and strategies to mitigate treatment-related toxicity.

The agent selected to achieve androgen deprivation should also be considered when trying to mitigate CVD risk, as the phase 3 HERO trial (NCT03085095) showed that relugolix, a novel oral gonadotropin-releasing hormone antagonist, achieved superior testosterone suppression and lower risk for major adverse cardiac events compared with leuprolide.⁶⁶ The REVELUTION-2 trial (NCT05320406) is an ongoing phase 3/4 open-label trial that is comparing cardiac outcomes between relugolix-abiraterone acetate and leuprolide-abiraterone acetate in hormone-naïve patients with advanced prostate cancer who are receiving pelvic radiotherapy.

Panelists emphasized the importance of collaboration with primary care providers (PCPs), who are often more up-to-date on CVD management, and the need to educate PCPs about the effects of PC therapies on CV health. Collaboration or referral to a cardiologist or cardio-oncologist is ideal but access may be limited, particularly in community practice settings. The panel discussed ways to optimize workflow to provide lifestyle counseling in a time-efficient manner and optimize management of the competing risks between PC and CVD and the relative CVD risks of PC treatments. Referral to community services (e.g., Silver Sneakers for exercise for patients on Medicare) or registered dietitians who can provide guidance on plant-based nutrition can help promote a healthy lifestyle.

Psychological and cognitive effects of ADT

Prospective research has shown ADT treatment in patients with PC has been associated with modest cognitive decline and increased risk for dementia.^{67,68} Treatment with ARPIs with ADT has been associated with increased risk for cognitive and mental impairment, seizures, fatigue, and falls compared with SOC treatment.⁶⁹ The ACE study showed that mean composite cognitive outcomes were similar between abiraterone and enzalutamide after 6 months, but perceived cognitive ability, fatigue, and depression were poorer in patients taking enzalutamide.⁷⁰ They emphasize the importance of assessing cognition before and during treatment to assess whether a patient is experiencing cognitive decline.

Depression is also common among patients with PC, with an estimated prevalence of 15%–20%.⁷¹ Multiple biologic, psychological, and social variables contribute to depression in PC, and pharmacologic ADT has been associated with a further increased risk for depression.^{71,72} Depression is also associated with lower likelihood of pursuing definitive treatment and worse OS outcomes, highlighting the importance of early identification and treatment.⁷³ The panelists noted that the link between androgen blockade and cognition and mental health outcomes is difficult to study because of the heterogeneity and confounding factors associated with PC, patient comorbidities, and life circumstances.

Some panelists said that identification of sleep disorders is important because they can contribute to cognitive decline and depression in patients taking androgen-directed therapy. Sleep disturbances (e.g., poor sleep quality, sleep apnea, clinical insomnia) are highly prevalent in patients with PC but are not routinely assessed, highlighting the need to screen for sleep disturbances to optimize PC survivorship and quality of life among caregivers.^{74,75}

Role of diet, exercise, and social support

Greater consumption of red and processed meats has been linked to a small but significant increase in risk for PC.⁷⁶ Consumption of more plant-based and less animal-based food has been associated with a lower probability of elevated PSA and lower risk for total, lethal, and fatal prostate cancer.^{77,78} A systematic review that included interventional and observational studies showed that a plant-based diet was associated with improvements in PC-related outcomes, and a prospective cohort study showed better quality of life (including sexual function, urinary health, and hormonal symptoms) in patients with nonmetastatic PC who consumed more plant-based and less animal-based food.^{79,80} Panelists noted that although multiple studies show plant-based diets can be less expensive than animal-based diets, food deserts may limit access to fresh produce, and they recommend counseling patients on the option of frozen fruits and vegetables—that often have as many or more nutrients than fresh options.^{81,82}

Higher levels of physical activity after diagnosis have been associated with lower cancer-specific mortality in PC survivors and

moderate improvements in cancer-specific quality of life and fatigue.^{83,84} However, most Americans do not reach the recommendations in physical activity guidelines (150 minutes of moderate-intensity aerobic activity and two sessions of resistance training), making it important to counsel patients on ways to increase physical activity.⁸⁵ Smoking cessation is important, as cigarette smoking at the time of PC diagnosis has been associated with increased PC-specific mortality after controlling for demographics, Gleason grade, stage at diagnosis, and primary treatment.⁸⁶ Addressing social support is also important, as partnered patients with PC have better quality of life than unpartnered counterparts.⁸⁷

In conclusion, with the rapid evolution of management across the PC spectrum, many questions remain about the optimal use of biomarkers, diagnostic imaging, and current therapies, as well as managing the long-term adverse effects of treatment. These include ways to use molecular biomarkers and artificial intelligence-guided models to individualize therapy in clinical practice, leverage PSMA-PET to improve early detection and treatment of high-risk occult metastatic lesions while minimizing overtreatment of lower-risk disease, and develop rational therapeutic combinations throughout the disease continuum. Developing novel ways to target AR and AR-independent treatment strategies is needed for ARPI-resistant mCRPC, and more research is needed to clarify the effectiveness of these treatments in the context of the changing treatment landscape in earlier disease settings. Finally, the increasing life expectancy of patients with PC highlights the importance of implementing lifestyle interventions (e.g., plant-based diets, physical activity, and sleep health) and multidisciplinary management of long-term health effects of therapy. The BTG PC panelists will use these insights to plan future conferences and educational efforts and will reconvene in 2026 to formulate consensus recommendations.

AUTHOR CONTRIBUTIONS

Rana R. McKay: Conceptualization, methodology, supervision, writing—original draft, and writing—review and editing. **Benjamin L. Maughan:** Supervision, conceptualization, writing—original draft, and writing—review and editing. **Alicia K. Morgans:** Conceptualization, supervision, writing—original draft, and writing—review and editing. **Neal D. Shore:** Conceptualization, supervision, writing—original draft, and writing—review and editing. **Evan Y. Yu:** Conceptualization, writing—original draft, writing—review and editing, and supervision. **Ravi A. Madan:** Conceptualization, writing—original draft, writing—review and editing, and supervision. **Jacob E. Berchuck:** Conceptualization and writing—review and editing. **Bradley C. Carthon:** Conceptualization and writing—review and editing. **Steven E. Finkelstein:** Conceptualization and writing—review and editing. **Leonard Gomella:** Conceptualization and writing—review and editing. **Michael A. Gorin:** Conceptualization and writing—review and editing. **Andrew W. Hahn:** Conceptualization and writing—review and editing. **Stacy Loeb:** Conceptualization and writing—review and editing. **Vivek K. Narayan:** Conceptualization and writing—review and editing. **Daniel P. Petrylak:** Conceptualization and writing—review and editing. **Charles J. Ryan:** Conceptualization and writing—review and

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CONFLICT OF INTEREST STATEMENT

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author on reasonable request.

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